From Transparent to Opaque: Transforming Explicit Reasoning Models to COCONUT via Intermediate FFN Layer

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Abstract

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1 Introduction

The Chain of Continuous Thought (COCONUT) [Hao et al., 2024] explores the potential of the latent layer of reasoning, as an alternate to traditional explicit CoT [Wei et al., 2022] in both reasoning procedure or answering procedure. Its prior work has shown that the latent layer of continuous reasoning can reduce unnecessary statements, and focus on the core of the problem.

These model issued are trained from general-purpose language, being supervised to train from steps of reasoning procedure, to progressively increase its layers of latent reasoning. It's impractical due to lack of customized reasoning data, and the progressive training is significantly less convenient to explicit CoT. Hence, we provide a new approach to implement these latent reasoning layer—fine-tuning from explicit reasoning models.

In January, the DeepSeek team proposed a distillation-based method for implementing a reasoning model via reinforcement learning [DeepSeek-AI, 2025], and they applied the approach to the Qwen-2.5B [Yang et al., 2024] to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to implement the COCONUT on the small model, and find its potential in reasoning, compared to the original explicit reasoning model.

References

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